

App Dev Project Report

1. Student Details

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About Me: I am a recent graduate now a working professional in the field of data science and analysis. I don't have much experience in web development but I learned a lot while building this one.

2. Project Details

Project Title: MAD 2 Vehicle Parking App

Problem Statement:

To design and build a web-based application that has both backend and frontend functionalities for a Vehicle Parking System, assisting both Users and Admin.

Approach:

I built the app using flask for backend and vue for front end, while using sqlalchemy for database modeling. I also used redis for server caching and celery for backend scheduled jobs.

3. AI/LLM Declaration

I used ai to assist me in the creation of my model and its integration with the flask application, also made use of ai in some frontend code, to get help in designing.

In my entire code, implementation of ai is around 15%.

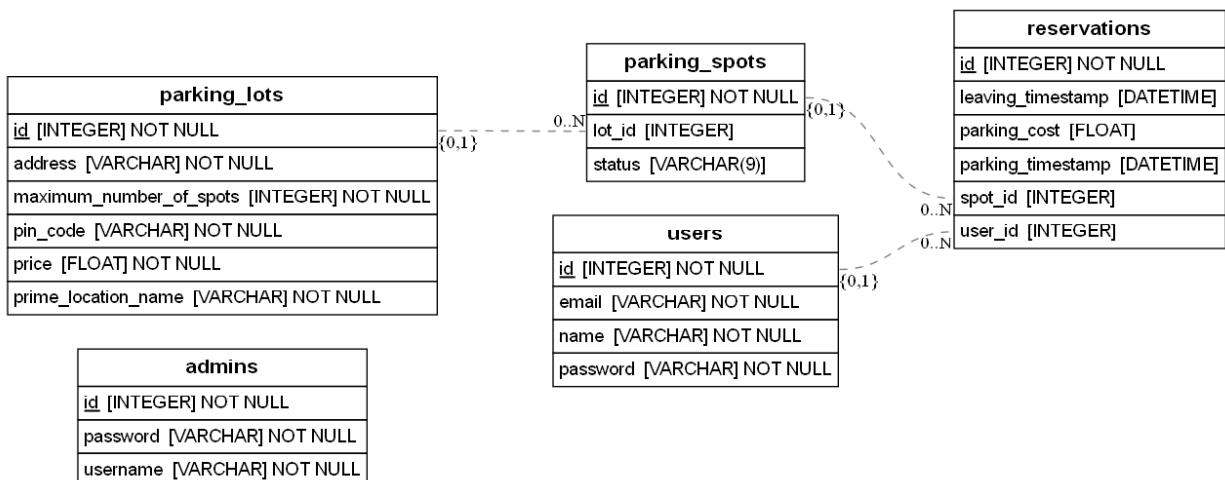
4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend code work
SQLAlchemy	ORM for database
VueJS	Frontend UI
Bootstrap 5	Frontend styling
Chart.js	Frontend charts
Redis	Caching
Celery	Backend scheduled jobs
SQLite	Storing user data

5. Database Schema / ER Diagram

Tables:

1. **Users** — for user details (id, name, email, password)
2. **Admins**— for admin details (id, password, username)
3. **Parking Lots**— for parking lots (id, address, max no. of spots, price,)
4. **Parking Spots**— for parking spots inside lots (id, address, max no. of spots, price)
5. **Reservations**— for user reservations (id, leaving time, parking time, spot id, user id)



6. Video Presentation

Drive Link:

https://drive.google.com/file/d/1T--Vpmh2u_ejiFsdwRAEczTiK9HMrbx/view?usp=sharing
